ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

MECHATRONICS

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Mechatronics paper 1 · practice a.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Use consistent accuracy. Accept other correct answers.

SECTION A

QUESTION 1

1.1 1.1.1 D ✓ 1.1.2 B ✓ 1.1.3 D ✓ 1.1.4 B ✓ 1.1.5 C ✓ 1.1.6 A ✓ 1.1.7 B ✓ 1.1.8 D ✓ 1.1.9 A ✓ 1.1.10 C ✓ (10)

1.2 1.2.1 True ✓ 1.2.2 False ✓ 1.2.3 True ✓ 1.2.4 True ✓ 1.2.5 True ✓ 1.2.6 True ✓ 1.2.7 False ✓ 1.2.8 True ✓ 1.2.9 True ✓ 1.2.10 False ✓ (10)

1.3 1.3.1 B ✓ 1.3.2 C ✓ 1.3.3 H ✓ 1.3.4 J ✓ 1.3.5 D ✓ 1.3.6 F ✓ 1.3.7 A ✓ 1.3.8 I ✓ 1.3.9 G ✓ 1.3.10 E ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 Any FIVE in a logical order, name ✓ + function ✓: compressor – compresses air; air receiver (tank) – stores air, smooths pressure, cools air; air dryer / after-cooler – removes moisture; service unit (filter, regulator, lubricator) – cleans air, sets the working pressure, adds oil mist; pipes / hoses and fittings – carry air; directional control valve – directs air to the cylinder; flow control valve – controls speed (10)

2.2 $A = \pi \times r^2 = \pi \times 0{,}025^2$ ✓ $= 0{,}00196\ m^2$ ✓; $F = p \times A = 600\ 000 \times 0{,}00196$ ✓ $= 1\ 178{,}10\ N$ ✓ (4)

2.3 A two-pressure (AND) valve ✓ – the cylinder extends only when both buttons are pressed ✓, so both of the operator's hands are away from the moving part ✓. (3)

2.4 The controller energises solenoid Y1 of the 5/2 valve, so air enters the rear of the cylinder and it extends ✓✓; when the piston magnet reaches the front reed switch, the switch closes ✓ and the controller energises solenoid Y2 (or de-energises Y1 with a spring return) so the cylinder retracts ✓; the rear reed switch signals that the cylinder is fully retracted and the cycle can repeat ✓. (5)

[22]

QUESTION 3

3.1.1 $A_{small} = \pi \times 0{,}01^2$ ✓ $= 0{,}000314\ m^2$ ✓; $p = F \div A = 200 \div 0{,}000314$ ✓ $= 636\ 619{,}77\ Pa \approx 636{,}62\ kPa$ ✓ (4)

3.1.2 $F_{large} = F_{small} \times (D \div d)^2$ ✓ $= 200 \times (100 \div 20)^2$ ✓ $= 5\ 000\ N$ ✓ (3)

3.1.3 The same volume of oil moves from the small cylinder to the large one ✓; the large piston has a bigger area, so it moves a smaller distance (force is gained at the cost of distance) ✓. (2)

3.2 Advantages (any THREE) ✓✓✓: very large forces; precise, smooth control of speed and position; oil is incompressible so it holds position under load; compact for the force produced; self-lubricating. Disadvantages (any TWO) ✓✓: oil leaks are messy and harmful; expensive components; fire risk with some oils; slower than pneumatics; needs return lines and a reservoir (5)

[14]

QUESTION 4

4.1 $\mathrm{n} = 120 \times 50 \div 4$ ✓✓ $= 1\,500\ \mathrm{r/min}$ ✓ (3)

4.2 $\mathrm{Slip} = (1\,500 - 1\,440) \div 1\,500 \times 100$ ✓✓ $= 4\%$ ✓ (3)

4.3 Start button (normally open) – energises the contactor coil, which closes the main contacts ✓; a holding (latching) contact keeps the coil energised after the button is released ✓; stop button (normally closed) – breaks the coil circuit and stops the motor ✓; overload relay – trips and opens the control circuit if the motor draws too much current for too long, protecting it from overheating ✓ (4)

4.4 Any TWO: speed can be adjusted to production needs ✓; soft starting reduces mechanical shock and starting current ✓; saves energy; can reverse the motor; built-in protection (2)

4.5 $\mathrm{P} = 2\pi \times 1\,440 \times 15 \div 60$ ✓✓ $= 2\,261{,}95\ \mathrm{W} \approx 2{,}26\ \mathrm{kW}$ ✓ (3)

[15]

QUESTION 5

5.1 (a) photoelectric (through-beam / retro-reflective) or capacitive sensor ✓ – detects non-metal objects ✓ (b) ultrasonic level sensor ✓ – measures distance to the water surface without touching it ✓ (c) load cell ✓ – converts weight into an electrical signal for an accurate mass ✓ (6)

5.2 Slip: belt can slip (except timing belts) ✓ – chain does not slip ✓; noise: belt is quiet ✓ – chain is noisier ✓; maintenance: belt needs tension checks, no lubrication ✓ – chain needs regular lubrication and tension adjustment ✓ (6)

5.3 Any SEVEN ✓✓✓✓✓✓✓: drain water from the receiver and filter; check the air pressure on the regulator; check for air leaks (listen, soapy water); top up the lubricator oil; inspect hoses and fittings for damage; check that sensors and reed switches are clean and secure; test emergency stops and guards; lubricate moving parts; check electrical connections and cables; check cylinder mountings and seals (7)

[19]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE